

Code: BCA 2004T (For Theory) BCA 2004P (For Practical)	SEC-II	Object Oriented Programming Using Java	3L+T: 4P	5 Credits (45 hours theory and 60 hours practical)
Max Marks; Theory: 100 (Int: 25; Ext: 75); Practical: 100				

Course Outcomes: Upon completion of the course, students will be able to

CO1: Understand the fundamental concepts of Object-Oriented Programming (OOP) and Java language structure.

CO2: Apply decision-making, branching, looping, and operator usage in Java programming.

CO3: Implement concepts of classes, objects, inheritance, polymorphism, arrays, and strings in Java programs.

CO4: Develop Java programs using packages and exception handling mechanisms.

Unit	Topics	Purposed lectures
I	Fundamentals of Object-Oriented Programming – Basic Concepts of OOP, Benefits and Applications of OOP. Java Evolution – Java Features, Difference between Java, C and C++, Java and Internet, Java Environment. Overview of Java Language – Setting up Java Development Environment (JDK, IDEs), Introduction to Simple Java Program, Use of Comments and Math functions, Application of two classes, Java Program Structure, Java Tokens and Statements, Implementing Java Program and JVM, Command Line Arguments.	11
II	Constants, Variables and Data Types – Constants, Variables, Primitive & Non-Primitive Data Types, Declaration of Variables, Giving values to Variables, Symbolic Constants, Typecasting. Operators & Expressions – Arithmetic, Relational, Logical, Assignment, Increment & Decrement, Conditional, Bitwise operators, Evaluation of Expressions, Type Conversions, Operator Precedence & Associativity. Decision Making, Branching & Looping – if-else, switch-case, for, while, do-while, Jump statements, Labeled loops.	11
III	Classes, Objects and Methods – Java Keywords, Defining Class, Instance Variables & Methods, Creating Objects, Methods Declaration, Constructors, this keyword, Static Members. Arrays, Strings and Vectors – 1D & 2D Arrays, Strings, Vectors, Wrapper Classes, Enumerated Types. Inheritance – Access Modifiers, Extending Classes, Interfaces, Multiple Inheritance using Interfaces, Polymorphism (Method Overloading & Overriding).	11
IV	Packages – Basics of Packages, System Packages, Creating User-Defined Packages, Creating and Accessing Packages, Adding Class to a Package. Exception Handling – try, catch, throw, throws, finally, Nested try, Multiple catch, Creating User-Defined Exceptions.	12

Text Books

1. Balaguruswamy, E. Programming with JAVA: A Primer. 7th ed. India: McGraw Hill Education, 2023.
2. Schildt, H. Java: The Complete Reference. 12th edition. McGraw-Hill Education, 2022.
3. Programming in Java 2, By Somasundaram, K, Jaico Publishing House.

Reference Books

4. Liang, Y. Daniel. Introduction to Java Programming. 7th ed., Pearson, 2008.
5. Malhotra, S., and S. Choudhary. Programming in Java. 2nd ed., Oxford UP, 2014.
6. Ivon Horton, Beginning Java-2, SPD Publication4
7. Ramesh Bangia, Learning JAVA-2, Khanna Book Publishing Co (P) Ltd